

Structural Influence: Measuring the Hidden Wiring of an NBA Offense

Rami Zheman (rami.zheman@gmail.com)
Independent Sports Analytics Researcher, Alexandria, VA

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Abstract

NBA teams spend billions on players, yet some of the hardest to characterize are the ones who shape how an offense operates. Box scores barely see them. Existing analytics measure production, efficiency, and usage, but not how an offense is organized or how much its structure changes around any one player. This paper maps teams' offensive organizations as a network, representing players and the play types they initiate. A player's structural influence is measured by how much the network's structure changes when he is removed, capturing how much the offense reorganizes in his absence. Comparing structural influence with usage identifies low-usage players whose absence can reorganize an offense as substantially as the loss of a high-usage playmaker.

Using public data across three NBA seasons, this paper finds that a player's organizational influence, after accounting for how much he touches the ball, is largely uncorrelated with conventional value metrics such as net rating. These players, referred to in this paper as shapers, instead affect the team's offensive footprint. On the floor, shapers are associated with teams running more cuts and pick-and-rolls and fewer spot-ups and drives. Their structural influence also partly persists after a trade, affecting the offensive organization of a new team. High-usage organizational talent reliably affects scoring efficiency; a shaper's structural influence does not. What an offense reorganizes around structurally is not the same as what makes it score.

This framework provides a way to study how an offense reorganizes when key players are unavailable, helping researchers examine offensive adaptation, coaches prepare for in-game offensive pivots, and front offices evaluate how a trade or absence may change team identity. Offensive organization becomes measurable, rather than solely a matter of coaching intuition.

1 Introduction

A motivating problem. NBA teams spend a lot of resources evaluating players, but some of the hardest guys to evaluate are the ones whose value is mostly organizational. Coaches, teammates, and front offices talk all the time about “glue guys” who connect an offense or hold a system together, even when their box score numbers look like a rotation player, not a difference-maker. At the February 2024 trade deadline, Dallas gave up a first-round pick for Daniel Gafford, a backup center averaging around eleven points and eight rebounds, and made the Finals with him starting at center. That trade is a clean way to ask the question this paper is actually about: *what was Dallas buying that production, efficiency, and usage numbers don't capture?*

There's already plenty of analytics work on how much a player contributes to winning [4, 5], and on what type of player he is [1, 3]. None of it really gets at how an offense is organized, or how dependent that organization is on one specific player. This paper doesn't prove that glue guys are secretly stars, and despite a lot of testing, it doesn't find that they add offensive efficiency at the

team level either. What it does instead is ask whether offensive organization can be measured at all, whether some types of players affect that organization more than others, and whether teams adapt when that organization changes.

Why this is worth measuring. Offensive organization is one of the last parts of team performance that’s still mostly qualitative, both for individual players and for teams as a whole. A reproducible measure of that organization, what this paper calls “wiring,” opens up questions existing value metrics don’t touch:

1. how offenses reorganize after an injury or absence, and how well they adapt;
2. which offenses are structurally *brittle*, leaning on one or two players, versus distributed and resilient;
3. how organizational responsibility shifts as young players develop;
4. whether a trade preserves a team’s offensive identity or rewires it;
5. which defenders disrupt an opponent’s organization the most.

None of these questions prove a player is worth some fixed number of extra points. They’re questions about structure, and depending on what you’re asking, they can surface real strengths or weaknesses in how a team is put together. As it turns out, that structure is largely independent of scoring efficiency (Sec. 6.4). What a front office is actually buying, in cases like Gafford, looks more like organizational continuity than points. Treating organization as something measurable and testable, instead of something you have to take a coach’s word for, opens up a set of questions basketball analytics hasn’t had a way to ask. The contribution of this paper is the framework itself and the measure it produces. What follows is a first pass at testing both, focused on one category of player in particular, not a broad survey.

Contribution A: the measurement framework. This paper introduces a reproducible, graph-based way to measure offensive organization, and how it reorganizes when a player is removed. A team’s offense is mapped as a graph connecting players to the play types they initiate. “Wiring” is the team-level map of which players jointly hold which offensive jobs. The test is what happens when a player is removed from that graph: his *structural influence* is how much the map changes without him. Crossing structural influence with usage isolates a group this paper calls SHAPERS: low-usage players with outsized residual influence on how the offense is organized. This is the offensive version of the glue guy.

Why a graph, specifically? It isn’t the easiest or most explainable tool available for this kind of problem, but it’s a natural fit for organizational relationships, which is a different goal than measuring outcomes. The question here isn’t how good an offense is, it’s how an offense’s organization changes when a player is taken out of it. A graph lets you remove one node and watch how the rest of it reorganizes, which is exactly the mechanism this paper needs.

Contribution B: what the framework shows. Applying the framework to three NBA seasons (2022–23 through 2024–25) surfaces a few things. There’s a real category of low-usage players who carry as much organizational weight as star playmakers (Sec. 6.1). That influence isn’t just value in disguise, it’s statistically separate from net rating and other value measures (Sec. 6.2). It shows up on the floor, not just in the math, shifting a team’s play-type mix when these players are in

the game (Sec. 6.3). It doesn't, however, translate into more points: having a SHAPER alongside a scorer and an organizer doesn't raise efficiency beyond what talent already explains (Sec. 6.4). It does still change team identity even when scoring stays flat, removing a SHAPER reorganizes a team's offense about as much as removing a star organizer does, more than removing anyone else (Sec. 6.4.2). Part of this influence travels with the player when he changes teams (Sec. 6.5), though predicting how well it'll travel in advance doesn't really work, aside from a weak and unconfirmed signal around cutting versus driving (Sec. 6.6).

Same play-type menu, different wiring. Figure 1 shows two teams, Minnesota and Houston in 2023–24, that ran nearly identical distributions of play types (menu cosine 0.99) but assigned responsibility for those play types very differently (wiring cosine 0.54). Minnesota leans on players who combine post-ups and spot-ups, or post-ups and pick-and-roll, more of an inside-out offense built around interior gravity. Houston leans on players who combine cuts, drives, and spot-ups, a more movement-based attack that uses cutting and driving to create the openings it then attacks. The team-wide totals for each play type end up nearly the same either way, because a player who mostly does one thing still adds to that play type's total count, he just doesn't contribute much to the overlap between play types. Two teams can look identical on a simple count of what they run and still be organized in completely different ways underneath. What this framework actually measures is how much that pattern of overlapping responsibility shifts when a given player is removed. SHAPERS are the focus of this paper because their structural influence is large relative to how much they touch the ball or show up in a box score.

Same menu ($\cos = 0.99$), different wiring ($\cos = 0.54$).
Top: Wolves green = more similarity in initiators for MIN; Rockets red = more similarity in initiators for HOU; white = teams alike.
Bottom: nearly identical play-type shares — the menu matches; the wiring does not.

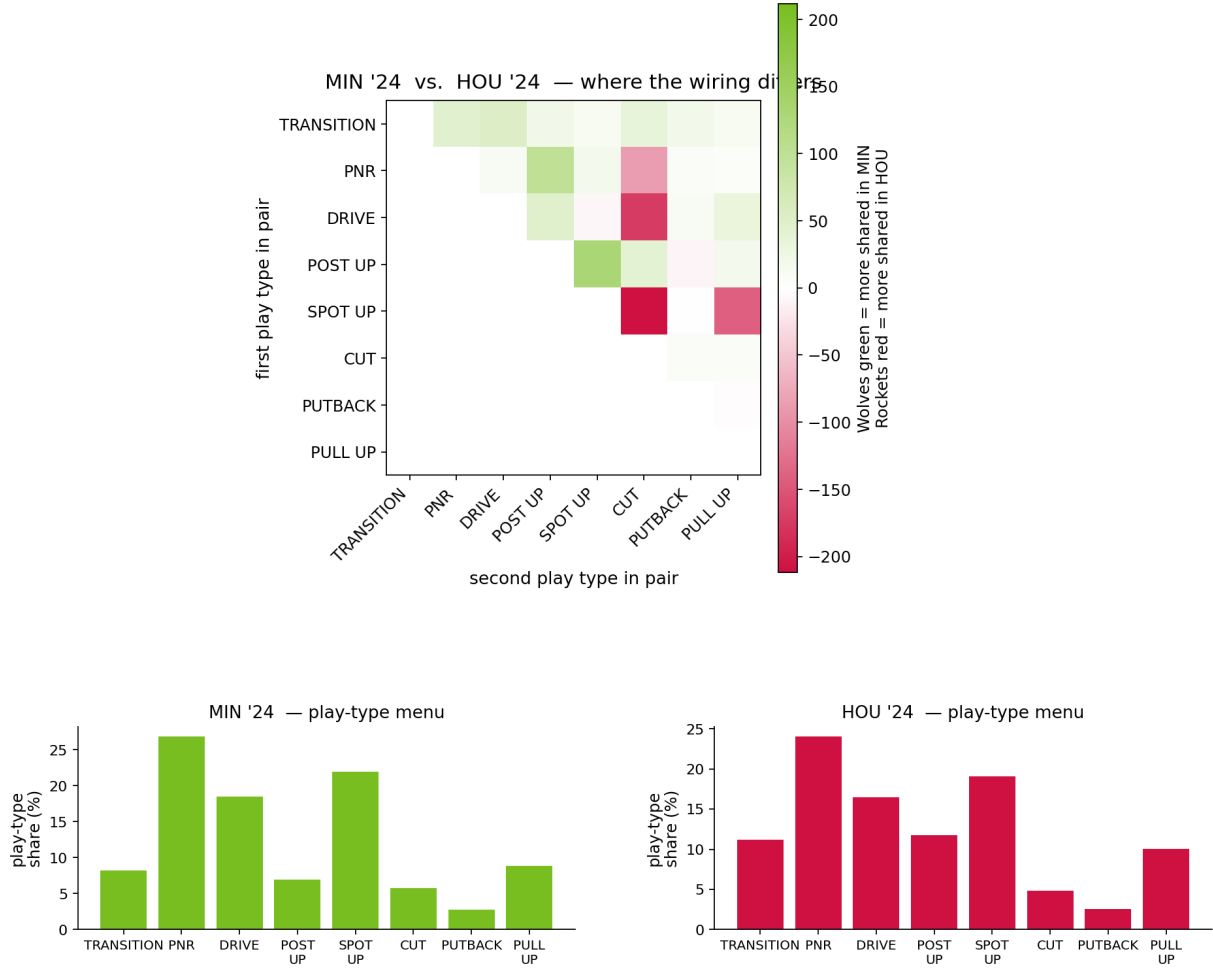


Figure 1. Same menu, different wiring. (1) Bottom bars = the menu. How often each team *initiates* each play type (marginal frequency). Minnesota (MIN) and Houston (HOU) are nearly identical (cosine 0.99). **(2) Top heatmap = where the responsibility maps differ.** Each cell is a play-type pair: shared-initiator score for Minnesota minus Houston (each first centered on its own team average). **Wolves green** = more initiator overlap for Minnesota; **Rockets red** = more for Houston; white $\approx$ teams alike. **Example:** POST_UP–SPOT_UP is green (Minnesota’s post-up and spot-up initiators overlap more); DRIVE–CUT and SPOT_UP–CUT are deep red. Same menu, different roster map.

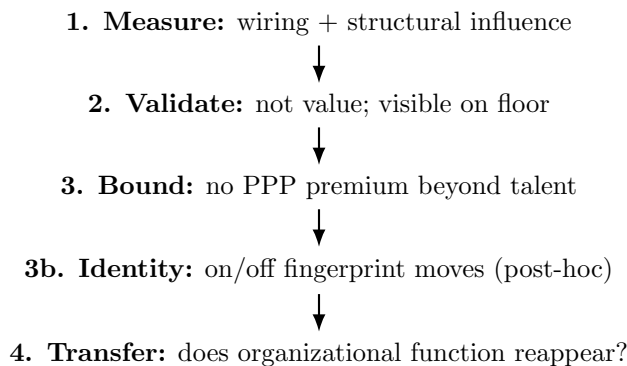


Figure 2. Paper spine. Contribution A builds the framework. Contribution B tests the framework and asks what it shows: distinctness from value, behavioral correspondence, an efficiency boundary, a disclosed post-hoc identity follow-up, and imperfect portability (Secs. 4–6.6). Shapers are the visible stress-test subclass, not the paper’s sole object.

Central claims.

1. Structural influence can be measured from initiation structure alone.
2. It isn’t the same thing as player value.
3. It travels with a player imperfectly when he changes teams.
4. Players carry organizational roles distinct from talent that show up in how an offense is shaped.
5. Different players can hold different combinations of responsibility even on teams with similar overall play-type distributions.
6. That influence is partly tied to the player himself, not entirely dictated by team context.
7. SHAPERS make this otherwise-hidden influence visible.

2 Related Work

Positions and archetypes. There have been various efforts to re-organize the five positions using box-score statistics and shot locations into data-driven “archetypes” [1, 3]. These describe what a player does, but not how that player’s actions relate to the actions of other players on the team. They also do not ask whether that role follows the player when he changes teams. In this paper, organizational function is defined by relational displacement (Sec. 4), and player portability is tested by looking at what happens when players move to a new team.

Player similarity and embeddings. Player embeddings can be used to find players who are similar based on their data [5]. This is useful for describing similarity between players, but it does not address whether a player’s organizational role carries over to another team.

Lineup and value models. Plus/minus models and net rating measure player and lineup performance and identify which groups perform well [4, 5]. They are designed to measure value, not how an offense is organized or whether a player’s organizational role can carry to another team. This paper measures something different. Sec. 6.2 and Sec. 6.4 show that structural influence does not add scoring efficiency beyond what existing measures of talent already explain.

Network analysis in basketball. Graphs have been used to model basketball possessions and calculate flow statistics, including work on the “price of anarchy” [2, 6]. Previous network studies show that graphs can be useful for studying basketball, but this work focuses on possession-level flow and team efficiency rather than measuring how individual players affect the structure of an offense.

The gap. Existing work can tell us about player types, player similarity, player value, and how the ball moves through a possession. It does not directly examine an offense as a map of player responsibilities: which players combine different play types, which players have an outsized effect on that structure, and whether that effect follows the player when he changes teams. That is what this paper tests.

3 Data

Source and unit. The data are possession-level records for three NBA regular seasons (2022–23, 2023–24, 2024–25). The atomic unit is a *possession*, with additional context like the initiating player, the play type, players on the court, the points scored, and other contextual fields like period, clock time, and score margin. After restricting to possessions with a clear initiator and a play type, the full dataset includes approximately 6.5×10^5 possessions across 90 team-seasons.

Play types. Each possession is labeled with one of $K = 8$ initiating play types (pick-and-roll, drive, post-up, spot-up, cut, putback, pull-up, and transition). The play type describes how a possession is initiated, reflecting the level at which offensive organization operates. Appendix H documents how the description-based inferrer derives labels from public NBA Stats play-by-play fields (actions, descriptions, assist flags, and shot distance/zone), including foul treatment.

PNR under-detection and calibration. The NBA public play-by-play data does not directly identify pick-and-roll (PNR) actions, so the description-based inferrer detected far fewer PNR possessions than expected. This matters because PNR is one of the play types used in the association and structural-influence calculations. To correct this, the PNR labels were calibrated to bring each team-season into a consistent PNR range of roughly 18–20%. The calibration promotes selected pull-up and unclassified possessions to PNR using player-level information (e.g. shot distance), rather than applying the same adjustment to every player. The calibrated labels, not the underlying proxy, are used in all association and structural-influence calculations. Applied consistently, the resulting PNR rates were 18.3–20.0% across seasons.

Initiators. The *initiator* is the first offensive player credited with a shot, turnover, foul, or free throw on the possession. For most play types this coincides with the player who begins the action (the driver on a drive, the shooter on a spot-up, the post player on a post-up, the cutter on a cut). On an assisted pick-and-roll, however, the initiator is the player who finishes the possession (the ball-handler on a pull-up or floater, or the roll man on an assisted rim finish) rather than exclusively the ball-handler who set up the action. This paper graphs this terminal-actor relation, rather than passing or assist credit, because it is the act that determines which player sets each possession’s structural context.

Seasons, filtering, and pools. For role assignment this paper defines a *rotation pool* of players with at least 15 minutes per game over at least 30 games in a team-season, ensuring each player’s initiation distribution is estimable. Players outside the pool are still classified but are excluded when computing distributional thresholds (Section 4.5). On-court composition analyses (Section 6.3) retain possessions with exactly five identified offensive players (99.7% of eligible possessions). The shaper-tier subgroup split in Sec. 6.4.1 was motivated by a review of 2025–26 data. After initial analysis, 2025–26 was identified as lacking full on-court coverage, which was subsequently corrected (Appendix G lists that season’s shaper roster). The check itself runs on the pre-registered three-season sample only.

4 Measuring Offensive Wiring

This section builds the framework (Contribution A): the offensive interaction graph, its play-type association matrix, and per-player structural influence, calculated as a network perturbation. From these calculations, role categorizations emerge, and shapers become visible through outsized displacement relative to usage. Notation is summarized in Table 1.

4.1 Nodes, edges, and the initiation matrix

For one team in one season, let $\mathcal{P} = \{1, \dots, n\}$ be the group of rotation players and let $\mathcal{K} = \{1, \dots, K\}$ be the group of the $K = 8$ play types. The offense is represented as a graph with two kinds of nodes, players and play types, where the only connections that exist are between a player and a play type, never player-to-player. Each connection has a weight: how many times that player initiated that play type. Collecting all of these weights gives an *initiation matrix* $\mathbf{B} \in \mathbb{R}_{\geq 0}^{n \times K}$, with one row per player and one column per play type, where each cell $\mathbf{B}[p, k]$ is simply the count of type- k possessions that player p initiated. Each play type’s column is then converted into a percentage breakdown across players, so instead of raw counts, each column shows what share of that play type’s initiations belongs to each player, and those shares for a given play type add up to 100%:

$$\tilde{\mathbf{B}}[p, k] = \frac{\mathbf{B}[p, k]}{\sum_{q \in \mathcal{P}} \mathbf{B}[q, k]}, \quad \sum_p \tilde{\mathbf{B}}[p, k] = 1 \quad \forall k. \quad (1)$$

4.2 The association matrix (min-overlap co-initiation)

This measure isn’t about how often each play type happens on its own, that’s a simple count anyone could pull from a possession table. What matters is whether the same players are behind two different play types, or whether separate specialists handle each one, since a possession table alone can’t tell you that.

The distinction matters because of what initiation represents: whoever initiates a play type is the player the offense currently depends on to produce that action. When two play types are held by the same players, high overlap, those actions function as a linked system. The offense can move between them without changing personnel, and a defender guarding that player has to respect both threats at once, since either could come from him. When two play types are held by entirely separate specialists, low overlap, the offense has divided those responsibilities into distinct jobs. No single defender has to guard for both, because no single player represents both threats. Across all 28 pairs, this pattern shows how responsibility is organized across a roster: which actions are combined in the same players, harder to defend and harder to remove without consequence, and which are divided across specialists, more easily replaceable. That’s what this paper means

by an offense’s *wiring*. It’s also why removing a player and watching which pairs collapse reveals something a raw initiation count can’t: not just what a player does, but which combinations of team responsibility existed only because he personally held both halves. See Appendix A for what the play-type pairs represent for basketball strategy and tactics.

Why certain pairs cluster in the same player is itself a basketball mechanism, not an arbitrary artifact of the data. Some play types share an underlying defensive exploit. A drive collapses help defenders inward, which is exactly what creates space for a pull-up once the defender recovers, or a spot-up once a helper strays too far from his own man. Cutting works the same way from the off-ball side: pulling defensive attention toward the ball opens driving and cutting lanes elsewhere. A player who is genuinely live in both halves of a pair like this is attacking the same defensive vulnerability two different ways, and a defender guarding him has to respect both threats at once. That’s what makes certain combinations, drive and pull-up, cut and drive, show up as real, above-chance links in the association matrix, rather than arbitrary pairs like putback and pull-up. See Appendix A for a full accounting of what each pair represents on the floor.

The overlap between two play types is the sum, across players, of the smaller of that player’s two initiation counts (computed on the *raw* initiation counts $\mathbf{B}$, not the column-normalized $\tilde{\mathbf{B}}$):

$$O[i, j] = \sum_{p \in \mathcal{P}} \min(\mathbf{B}[p, i], \mathbf{B}[p, j]), \quad i, j \in \mathcal{K}. \quad (2)$$

The minimum matters because a player only contributes strongly to the link if he does meaningful volume in both play types. A player with 100 cuts and 5 spot-ups contributes only 5. A player with 60 cuts and 55 spot-ups contributes 55. The measure therefore rewards players who genuinely hold two jobs together rather than specialists who dominate only one. This is not a sequence measure within a possession. It asks whether the same players show up doing both jobs across the season.

That team-level number still has a problem: a star player who initiates a lot of every play type, a Nikola Jokić or LeBron James, will show up in a lot of these overlap totals just because he touches the ball more, not because any two of his play types are meaningfully linked. To correct for that, each pair’s raw overlap number gets compared against a random baseline built from shuffled versions of the same team. To separate real overlap from what would occur simply because some players initiate more possessions than others, each team’s initiation matrix is compared with 1,000 randomized versions. The randomization preserves every player’s total initiation volume and each play type’s total team frequency, while randomly reallocating play-type labels across the team. The observed overlap for each play-type pair is then standardized against its own shuffled distribution. That standardized distance, rather than the raw overlap count, is what is recorded for each pair. With $\mathbb{E}_0[\cdot]$ and $\text{sd}_0[\cdot]$ the null mean and standard deviation,

$$\mathbf{A}[i, j] = \frac{O[i, j] - \mathbb{E}_0[O[i, j]]}{\text{sd}_0[O[i, j]]}. \quad (3)$$

Those 28 standardized values together are what this paper calls a team’s *structural fingerprint*, a team-level object throughout. Players only enter the calculation as building blocks summed into $O[i, j]$, never as a separate fingerprint of their own. A team looks more like itself, comparing one half of its season to the other, than it looks like any other team, and that stays true even after removing how often each play type is simply run. That’s the basis for treating this fingerprint as a real, stable property of a team’s offense, not just a snapshot of one season’s noise.

4.3 Structural influence: how much the offense’s wiring changes when a player is removed

This paper defines a player’s structural influence by asking a simple question: what happens to a team’s fingerprint if he’s removed entirely? Take the full fingerprint for a team, built from every rotation player as described above, and call it $\mathbf{A}_{\text{full}}$. Now delete one player completely, remove every one of his initiations from $\mathbf{B}$, redo all of the shuffle and calculation processes for each play-type pair, and get a new fingerprint, $\mathbf{A}_{-p}$. Comparing those two fingerprints, before and after, is the idea. The difference between the full and leave-one-out fingerprints is measured with cosine similarity, which compares their pattern rather than their overall magnitude. Structural influence, $L(p)$, is defined as:

$$L(p) = 1 - \cos(\text{vec}(\mathbf{A}_{\text{full}}), \text{vec}(\mathbf{A}_{-p})) = 1 - \frac{\langle \text{vec}(\mathbf{A}_{\text{full}}), \text{vec}(\mathbf{A}_{-p}) \rangle}{\|\text{vec}(\mathbf{A}_{\text{full}})\| \|\text{vec}(\mathbf{A}_{-p})\|}. \quad (4)$$

A player therefore has high structural influence when removing him changes the pattern of play-type relationships, not simply when removing him reduces the number of possessions. A high-volume player whose absence shrinks the same pattern should not automatically receive high structural influence.

Figure 3 shows this using two Minnesota players from the same 2023–24 team, before any further adjustment of the measure. The heatmap shows the full, cell-by-cell change in the fingerprint, not just the single summary number $L(p)$. This color differentiation is defined later as the displacement signature (Section 4.6) but provides context for the following discussion.

For Rudy Gobert (usage 0.09, $L = 0.60$): removing him from the fingerprint collapses several cut-related links. CUT–PNR moves, consistent with his own game, rim runs and roll starts off ball screens, a real dual threat that forces a defense to guard both the roll and the backdoor cut. CUT–DRIVE, CUT–SPOT_UP, and CUT–PULL_UP move even more sharply, and those aren’t Gobert’s own combinations, they belong to other Minnesota players. Karl-Anthony Towns (usage 0.15, $L = 0.00$), on the same roster: removing him leaves the fingerprint almost completely unchanged, because other players were already covering the same combinations he touched.

A simple count shows cutting gets rarer without Gobert, Minnesota’s cut rate falls from 7.5% to 3.1% of possessions. But a count alone can’t say whether the cuts that remain are still paired with a drive, pull-up, or spot-up from the same player, or have become isolated actions nothing else backs up. That’s what the collapsing CUT–DRIVE, CUT–SPOT_UP, and CUT–PULL_UP pairs actually capture: fewer players cutting and driving or shooting at real volume, meaning fewer two-way threats a defense has to respect. A cutter who never drives or shoots off that same threat is easy to sag off. Stated as a hypothesis and not tested here, a defense can start treating a cut as a one-off help rotation instead of a real coverage responsibility once fewer players are running it as one half of a live threat.

That contrast, deep and structured for Gobert, shallow and sparse for Towns, is what structural influence is built to capture, something a raw count of cuts can’t show. This is a hypothetical, leave-one-out calculation based on removing a player from the season’s data, not something observed from an actual injury or game.

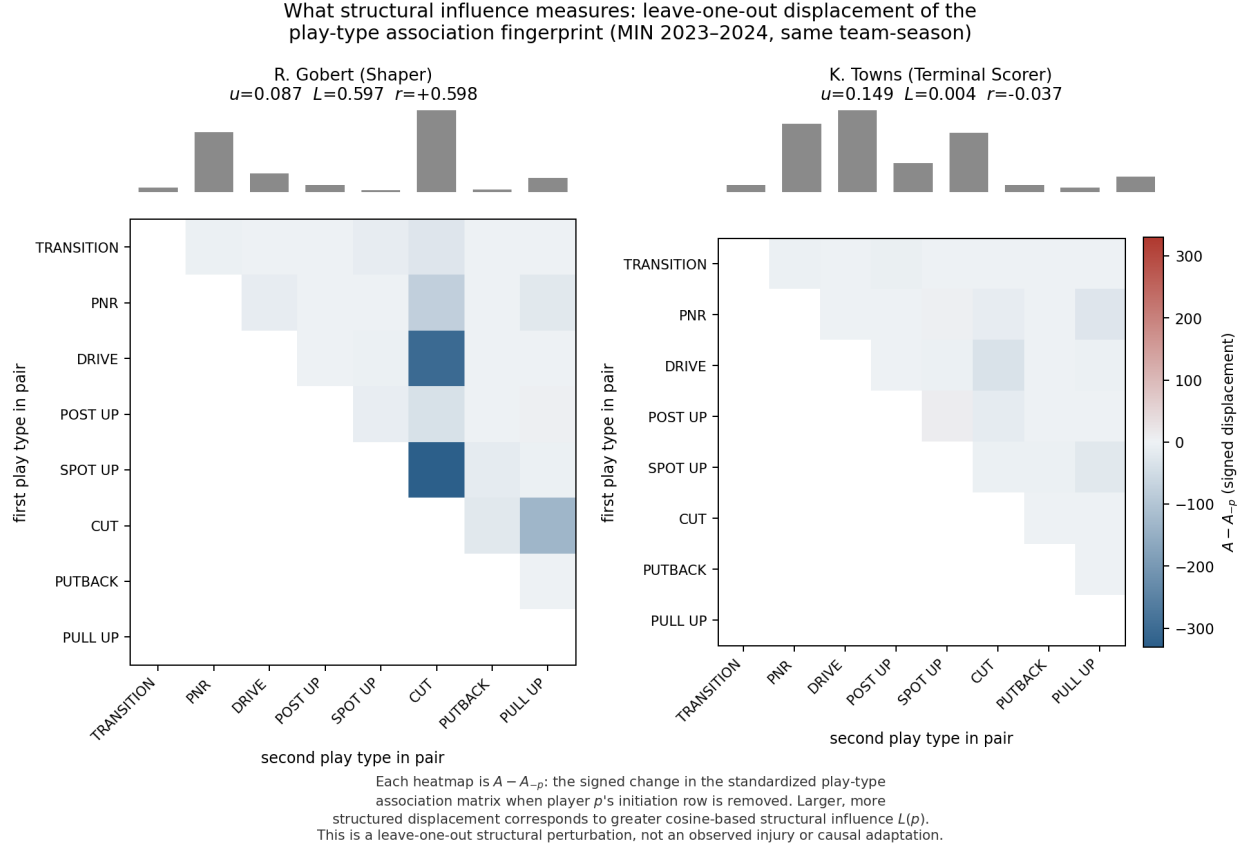


Figure 3. What structural influence measures: leave-one-out displacement of the play-type association fingerprint. Two players from the same team-season (Minnesota, 2023–24), same 8×8 play-type ordering, same diverging scale, diagonal blank. Each heatmap is $A_{\text{full}} - A_{-p}$, the signed change in the standardized play-type association matrix when player p 's initiation row is removed (the displacement signature of Sec. 4.6); the bar strip above each panel is p 's own play-type profile π_p , shown only for context. *Left:* Rudy Gobert (SHAPER, $u = 0.09$, $L = 0.60$, $r = +0.60$): removing him collapses several CUT-linked associations, a concentrated, deep displacement. *Right:* Karl-Anthony Towns (TERMINAL SCORER, $u = 0.15$, $L = 0.00$, $r = -0.04$) on the same roster: removing him leaves the fingerprint almost unchanged, a sparse, shallow displacement. The heatmap makes structural influence visible as the *pattern* (not merely the magnitude) of play-type association links displaced by a player's removal. This is a leave-one-out structural perturbation of the measured fingerprint, not an observed injury or causal adaptation.

4.4 Usage-residual structural influence

Structural influence, $L(p)$, still has one problem: it partly just reflects how much a player plays. Someone who touches the ball more will mechanically show up in more of these overlap calculations, even without any real connective role. To separate structural influence from volume, this paper compares each player's $L(p)$ against what you would expect purely from his usage rate, and keeps only the leftover. This paper fits a curve (allowing for some curvature, not just a straight line) predicting $L(p)$ from usage across every classified player. Whatever a player's actual $L(p)$ is above

or below what that curve predicts, given his usage, is called his residual influence, $r(p)$:

$$L(p) = \beta_0 + \beta_1 u_p + \beta_2 u_p^2 + r(p), \quad r(p) = L(p) - (\hat{\beta}_0 + \hat{\beta}_1 u_p + \hat{\beta}_2 u_p^2). \quad (5)$$

A positive $r(p)$ means a player's structural influence is higher than his usage alone would explain. A negative $r(p)$ means it is about what you would expect, or lower. One distinction worth being precise about: the usage used in this residual calculation is a graph-native measure, a player's share of his team's total initiations, calculated directly from the same data as everything else in this paper. That is different from the standard box-score usage rate reported elsewhere (like Table 4) and used later to build the taxonomy below. The two usually track each other closely, but they are not the same number, and keeping them separate means the residual is not just mechanically guaranteed to look clean against the box-score version.

4.5 Role categorizations: making shapers visible

With both usage and residual influence, players can be categorized into four groups by crossing the two values. The first cut is usage: is a player above or below the median usage of the rotation pool, meaning does he play a high-volume or low-volume role. The second cut is residual influence: is $r(p)$ positive or not, meaning does he carry more structural influence than his usage alone would predict. Four groups fall out of that:

$$\text{role}(p) = \begin{cases} \text{PRIMARY ORGANIZER} & u_p \geq u_{\text{med}} \wedge r(p) > 0, \\ \text{SHAPER} & u_p < u_{\text{med}} \wedge r(p) > 0, \\ \text{TERMINAL SCORER} & u_p \geq u_{\text{med}} \wedge r(p) \leq 0, \\ \text{ROLE OCCUPANT / SPECIALIST} & u_p < u_{\text{med}} \wedge r(p) \leq 0. \end{cases} \quad (6)$$

- **High usage, $r(p) > 0$: PRIMARY ORGANIZER.** A high-volume player whose structural influence exceeds even what his heavy usage predicts, someone whose presence organizes initiation pathways beyond just taking a lot of shots. These are usually playmakers.
- **Low usage, $r(p) > 0$: SHAPER.** This is the group this paper is built around: players who barely touch the ball but whose absence reorganizes the offense more than their usage would suggest.
- **High usage, $r(p) \leq 0$: TERMINAL SCORER.** A high-volume player whose structural influence is basically just what you would expect from that volume, nothing extra.
- **Low usage, $r(p) \leq 0$: ROLE OCCUPANT / SPECIALIST.** A low-volume player whose structural influence is also about what you would expect, nothing extra.

Most players land in the TERMINAL SCORER and ROLE OCCUPANT / SPECIALIST groups. That is expected because most players simply are not unusually structurally important relative to how much they play, and only a small number clear that bar in either usage bracket. Splitting the two high-usage groups apart matters, because it separates the organizers (players whose usage comes with real organizing responsibility) from simply high-volume scorers and finishers whose influence is just what their volume already explains. It is possible for both types to exist on the same roster at once, and this paper covers a real example of that later (Sec. 6.1). These four categories exist mainly to make SHAPERS visible as a clear category. The other three groups are not a separate research focus but could be. In this paper, they are documented for contrast against SHAPERS.

Table 1. Notation.

Symbol	Meaning
$\mathbf{B}, \tilde{\mathbf{B}}$	player $\times$ play-type initiation matrix; column-normalized form
$\mathbf{O}$	min-overlap co-initiation matrix, Eq. (2)
$\mathbf{A} = \mathbf{A}_{\text{full}}$	fixed-margin standardized association matrix (fingerprint), Eq. (3)
$\mathbf{A}_{-p}$	association matrix with player p removed
$L(p)$	structural influence = how much the offense’s wiring depends on p , Eq. (4)
$r(p)$	usage-residual structural influence, Eq. (5)
$\Delta\mathbf{A}_p \in \mathbb{R}^{28}$	adjacency-displacement signature (player’s structural footprint), Eq. (7)
$\boldsymbol{\pi}_p \in \Delta^{K-1}$	player’s play-type profile (table-visible)
u_p	usage rate

4.6 Displacement signature: measuring the portability of a player

Figure 3 already showed a cell-by-cell view for Gobert and Towns, rather than collapsing each player down to a single $L(p)$ number. That full picture, exactly which play-type pairs moved and in which direction when a player is removed, is what this paper calls the *displacement signature*, $\Delta\mathbf{A}_p$:

$$\Delta\mathbf{A}_p = \text{vec}(\mathbf{A}_{\text{full}}) - \text{vec}(\mathbf{A}_{-p}) \in \mathbb{R}^{28}. \quad (7)$$

Where $L(p)$ says how much the fingerprint moved overall, $\Delta\mathbf{A}_p$ says which specific pairs moved, and which direction. It records exactly which combinations of play types a given player is holding together. There is another way to describe a player that might sound similar but is not the same thing: his own play-type profile, meaning simply what percentage of his own initiations fall into each play type. That is something you could read directly off a normal stat sheet; it just describes what a player does. $\Delta\mathbf{A}_p$ is different: it is not about what he does, it is about what changes on the rest of the team when he is gone. Two players could have very similar play-type profiles and still have very different displacement signatures, because a profile describes the player, while the signature describes his structural influence on everyone else. This distinction matters later, in the portability test (Section 6.5), which asks whether a player’s own displacement signature carries over to a new team after a trade. To make sure that any apparent carryover is not just because he plays similarly wherever he goes, the test compares his signature against decoy players on the new team who share a similar play-type profile, so any leftover similarity in signature specifically, beyond just similar play style, can be credited to something more player-specific than just running the same plays.

5 Pre-registration

The portability and efficiency analyses are *pre-registered*. For each analysis, the methods were fixed before looking at results: a single hypothesis and outcome metric, a null model, a feasibility check confirming the test could actually be run, and stop conditions that locked the question and the data in advance, so nothing could be swapped out after seeing how the results looked. Because players are not independent of each other, players on the same team or in the same franchise tend to have correlated results, the statistical tests account for that clustering rather than treating every player-season as separate, uncorrelated data points. This is what makes each null trustworthy, not just a test that happened to come back empty because it was not run carefully.

6 Results

Contribution B, in order: does the measure reveal something new (Sec. 6.1); is it just value (Sec. 6.2); does it match observable basketball (Sec. 6.3); where does it stop (Sec. 6.4); does identity move when efficiency does not (Sec. 6.4.2, disclosed post-hoc); is it partly player-specific (Sec. 6.5); can transfer be predicted (Sec. 6.6).

6.1 Does the measure reveal something new?

Applying Eq. (6) to the rotation pool (924 player-seasons; the pre-registered three-season sample, 2022–23 through 2024–25) produces four disparate, differently sized categories as expected under a residual cut (Table 2). SHAPER players have structural influence (0.078) comparable to PRIMARY ORGANIZERS (0.094) while initiating far fewer possessions (usage 0.145 vs. 0.244), and scoring less than half as much (9.0 vs. 18.4 ppg). Their residual influence (+0.062) is the highest of any category. TERMINAL SCORERS are the mirror image: high usage (0.233), real scoring (16.6 ppg), near-zero residual influence (−0.019), which means they score about as expected from usage. Most of the pool sits in the two other categories (terminal 40%, specialist 39%). The graph identifies a type of structural importance that other offensive measures do not see: organizer-level influence on a team’s offensive wiring at about half the scoring volume.

Table 2. Organizational-function classes. Rotation pool ($n = 924$ player-seasons, 2022–23 through 2024–25). Usage is box-score usage rate (separate from the graph-native initiation share u in Eq. 5). Influence is $L(p)$ (Eq. 4); residual influence is $r(p)$ (Eq. 5). Categories use usage median $\times r > 0$ (Eq. 6). On-court net rating is omitted from this table and examined in Sec. 6.2.

Class	n	% pool	Usage	Influence	Resid. influence	PPG	MPG
PRIMARY ORGANIZER	96	10.4	0.244	0.094	+0.054	18.4	29.6
SHAPER	103	11.1	0.145	0.078	+0.062	9.0	23.6
TERMINAL SCORER	368	39.8	0.233	0.014	−0.019	16.6	28.6
ROLE OCCUPANT / SPECIAL.	357	38.6	0.147	0.006	−0.009	8.2	22.8

Table 3 and Figure 4 examine actual players against the structural influence and usage measures. The Organizer/Terminal split is not “who creates” versus “who scores,” because both classes take high usage and initiate a lot. It asks a more specific on-court question: when this player leaves the floor, does the team’s map of who starts which play types reorganize beyond what losing that much volume would imply?

Table 3. Named exemplars by category. One player-season each. Usage is box-score usage rate; $L = L(p)$; $r = r(p)$. Bold rows are labeled in Fig. 4. Illustrations, not a ranking.

Category	Player	Tm	Season	Usage	Infl. L	Resid. r
ORGANIZER	S. Gilgeous-Alexander	OKC	2023–24	0.317	0.308	+0.216
	A. Davis	LAL	2023–24	0.261	0.168	+0.105
	N. Jokić	DEN	2024–25	0.285	0.145	+0.067
	B. Adebayo	MIA	2023–24	0.247	0.211	+0.166
	D. Ayton	POR	2023–24	0.211	0.327	+0.305
	A. Sengun	HOU	2023–24	0.270	0.136	+0.093
SHAPER	R. Gobert	MIN	2023–24	0.152	0.620	+0.598
	D. Gafford	WAS	2022–23	0.146	0.411	+0.397
	J. Poeltl	TOR	2023–24	0.160	0.217	+0.203
	J. Allen	CLE	2024–25	0.156	0.110	+0.088
	W. Kessler	UTA	2023–24	0.127	0.239	+0.227
	J. Duren	DET	2022–23	0.139	0.240	+0.227
TERMINAL SCORER	L. James	LAL	2023–24	0.285	0.013	−0.054
	J. Tatum	BOS	2022–23	0.319	0.009	−0.090
	J. Brown	BOS	2022–23	0.307	0.023	−0.050
	F. Wagner	ORL	2023–24	0.250	0.011	−0.044
	C. McCollum	NOP	2022–23	0.260	0.035	−0.027
ROLE OCC./SPEC.	J. Hart	NYK	2023–24	0.133	0.003	−0.022
	M. Smart	BOS	2022–23	0.176	0.001	−0.020
	D. Finney-Smith	BKN	2022–23	0.129	0.005	−0.007
	O. Anunoby	NYK	2023–24	0.173	0.001	−0.011
	J. Holiday	BOS	2024–25	0.156	0.001	−0.017

How structural categories differ in offensive initiation. The graph does not track assists, within-possession passing chains, screens set without initiation credit, gravity, or communication. An initiator is as defined in Sec. 3. Category-average initiation play types make the roles more visible. PRIMARY ORGANIZERS start half-court offense on the ball: pick-and-roll ($\approx 24\%$), drives ($\approx 19\%$), and post-ups ($\approx 16\%$) lead. TERMINAL SCORERS look similar, where drives and pick-and-rolls also dominate, but they do more spot-up shooting and are less unique overlap across those jobs. Other players already start the same combinations, so removing the Terminal Scorer mostly just removes possessions rather than rewiring the offense. SHAPER players are typically not primary handlers. They focus on pick-and-roll ($\approx 25\%$) and cuts ($\approx 23\%$). ROLE OCCUPANTS / SPECIALISTS are mostly spot-up shooters ($\approx 33\%$ spot-ups). High residual influence (r) means the player’s initiation jobs are shared across play types in a way the rest of the players on the team do not mirror. This is the structural influence.

Player comparisons. The same seasons are bolded in Table 3 and labeled in Figure 4. On 2023–24 Oklahoma City, SGA is the on-ball organizer: drives $\approx 34\%$, pull-ups $\approx 24\%$, pick-and-rolls $\approx 18\%$ of his initiations, and he alone starts roughly a third of Thunder drives and pull-ups and nearly half their post-ups ($r = +0.22$). Denver 2024–25 is the same shape with Jokić as the half-court hub (pick-and-roll and post-up lead). He starts over half of Denver’s post-ups; $r = +0.067$.

By contrast, 2023–24 Los Angeles has two high-usage stars in opposite categories: the dotted link in Figure 4. LeBron (drives and PNRs at star volume; $r = -0.054$) is a **TERMINAL SCORER** for this Lakers roster: the coaching “point forward” label fits what he does with the ball, but with Davis already holding the post/cut/PNR overlap, those initiation jobs are ones the Lakers roster can actually redistribute. Anthony Davis (pick-and-roll, post-up, and cuts; $\approx 42\%$ of Lakers post-up starts and $\approx 31\%$ of their cuts; $r = +0.105$) is the labeled Organizer. He serves as the interior/ball-screen hub next to LeBron’s volume. Boston 2022–23 is the wing version of the same split: Tatum’s drive/PNR/spot-up volume with $r \leq 0$, while the motion offense keeps those play initiations shared.

The sharpest illustration is the low-usage half of the same split. Gobert and Gafford are not traditionally described as offensive organizers. They are not creators for teammates in the box-score sense. Yet they carry organizer-level structural dependence without the volume. Gobert alone started $\approx 62\%$ of Minnesota’s cuts in 2023–24 (cut and pick-and-roll $\approx 74\%$ of his own initiations), as shown in Section 4.3, not as a creator, but as a node the offense’s geometry depends on. That is what usage cannot see, and why shapers are the category of focus in this paper. Hart is a Role Occupant: mixed spot-up/PNR/drive volume that others also cover. The secondary table rows are additional in-class examples not labeled on the figure.

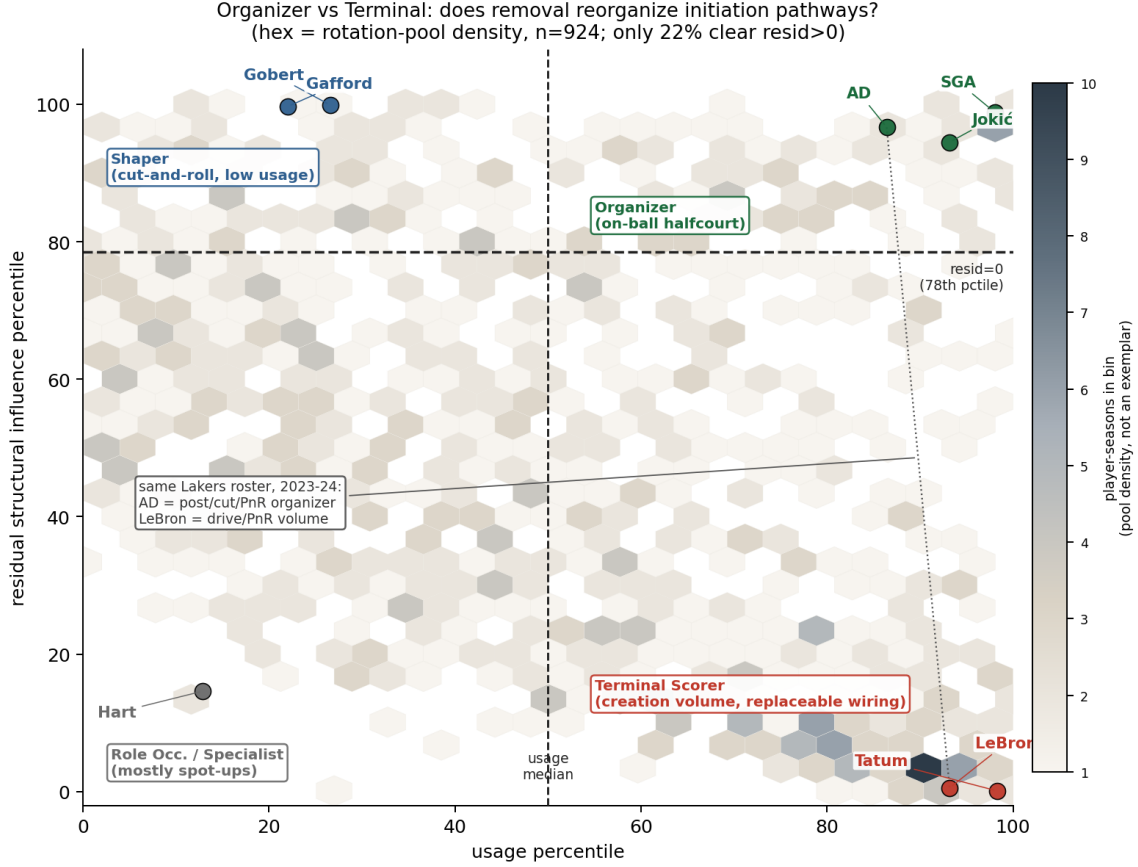


Figure 4. Organizer vs. Terminal on one map. Both axes are percentile ranks of the player pool ($n = 924$). Vertical dashed line: usage median. Horizontal dashed line: $r = 0$ (Eq. 6), at the ≈ 78 th residual percentile; only $\approx 22\%$ of the pool clears it. Bold rows in Table 3 are the labeled points. In basketball terms: SGA/Jokić/AD are high-usage half-court initiation hubs (PNR/post/drive overlap, $r > 0$). Gobert/Gafford are low-usage cut-and-roll hubs. LeBron/Tatum take star creation volume with $r \leq 0$. Hart is a replaceable Role Occupant. The dotted link is the within-roster illustration: on the 2023–24 Lakers, Davis holds the post/cut/pick-and-roll overlap while LeBron holds high-usage volume. Hexbin shading is pool density only (dark bin near Tatum = many Terminals, not one player).

6.2 Is it just another value metric?

One worry is that structural influence might not be measuring anything new at all, that it could just be another way of re-detecting usage, scoring, or how good a player’s team already is. To check this, this paper looks at whether SHAPERS tend to have unusually strong on-court net rating, once team quality is accounted for. If SHAPERS turned out to have a real, hidden edge in net rating, that would suggest structural influence is secretly just picking up on player value under a different name. If there’s no such edge, that rules out that explanation, and supports the idea that structural influence is measuring something distinct from value.

Comparing organizers and shapers as categorized above, before adjusting for team quality, organizers already beat shapers on raw net rating, $+1.45$ vs. $+0.80$. This leaves no hidden shaper premium to explain. Even after adjusting net rating for team quality, organizers still lead shapers, $+1.22$ vs. $+0.50$, but the gap does not clear a significance bar ($p = 0.10$), so this does not prove

Table 4. Structural influence vs. traditional metrics. Spearman ρ , $n = 1,283$ player-seasons with complete box-score coverage, broader than the 924-season player pool in Table 2. The key cell is $\rho(r, \text{net rating}) = -0.018$: residual structural influence has essentially no relationship with on-court net rating in this sample.

Metric	$\rho(L, \cdot)$	$\rho(r, \cdot)$
Usage rate	+0.270	-0.261
Points per game	+0.436	-0.264
Assists per game	+0.278	-0.255
Assist %	+0.106	-0.190
Touches	+0.376	-0.253
Frontcourt touches	+0.424	-0.238
Net rating (value)	+0.203	-0.018

the difference is real either.

Raw influence does correlate modestly with net rating, points, and touches, which is consistent with high-volume players appearing more often in the underlying overlap calculations. Once usage is removed, those relationships largely disappear. The residual measure therefore does not appear to be simply re-measuring volume, scoring, or on-court net rating. In this sample, organizational function and the conventional measures examined here are capturing different things.

6.3 Can we actually see what these shapers do on the court?

Having measured who the wiring depends on, and shown that structural influence is not just another way of measuring player value (Sec. 6.2), this paper checks whether the category corresponds to what we actually see on the court: the category-level version of the rim-running, cut-and-roll interior role shown in the Table 3 examples. Using within-team-season comparisons, this paper asks whether a shaper on the court changes the team’s play-type mix. That would corroborate that the category is visible on the floor, not just in the math.

Shaper presence and play mix. Across 73 team-seasons, a shaper on the floor shifts the mix toward off-ball and ball-screen actions and away from spot-up and drive actions: cuts +1.35 pp, pick-and-roll +0.98 pp; spot-ups -1.25 pp, drives -0.54 pp (percentage points of share, within-team). The category’s low usage is what makes this meaningful: shapers are low usage, so they are not taking many shots. The shift is better explained as reorganization among the other four players.

For comparison, primary organizers reshape the mix even more sharply (post-ups +2.33 pp, pick-and-roll +0.96 pp; 58 team-seasons). The team-season count differs between the two groups (73 vs. 58) because a team-season only enters this comparison if it has at least one on-court possession with that category present and at least one without. Which team-seasons clear that bar depends on the category, since organizer and shaper presence are not totally common across rosters.

These are associations, not causal estimates, for two reasons. First, high-usage categories mechanically run more of their own play types simply by being on the floor and taking those shots themselves; some of their mix shift reflects that, not real reorganization. This is part of why the shaper contrast is more informative: shapers barely touch the ball, so a mix shift when they are on the floor cannot be explained the same mechanical way. Second, the minutes when a shaper is off the floor often overlap with bench lineups and garbage time, situations where the competitive intensity and personnel are already different for reasons that have nothing to do with the shaper.

Comparing a team to itself removes differences in overall team identity, but it does not remove the fact that “shaper on” and “shaper off” minutes can come from genuinely different game situations, not just the same lineup with one player swapped out.

A refinement: rebalancing the offense, not expanding it. There are two ways a shaper’s presence could change the offense. He could be adding something genuinely new, more distinct kinds of actions that did not exist without him. Or he could simply be shifting how often the offense’s existing actions get used, without adding anything new at all.

The data supports the second explanation. Shaper presence does not make the offense’s play-type mix more varied or unlock actions that were not already part of the team’s play mix. Instead, the effect is narrower: with a shaper on the floor, the offense changes how often it uses actions it already has, leaning more toward cuts and PNR and less toward spot-ups and drives.

6.4 Where do we stop?

The framework so far measures organization, not scoring. A natural next question is whether it also predicts what matters most in practice: points per possession.

It didn’t hold. A measurement of how an offense is organized doesn’t have to predict how well it scores, and this one doesn’t. The result puts a useful boundary around the measure: structural organization is not a hidden efficiency metric.

How this was tested. Using every possession where all five players on the floor were classified into a role ($N = 4.2 \times 10^5$ possessions across 90 team-seasons), coverage is defined as an indicator that at least one organizer, one shaper, and one terminal scorer are all on the floor at once:

$$\text{COVERED}_i = \mathbb{1}[n_i^{\text{org}} \geq 1 \wedge n_i^{\text{shap}} \geq 1 \wedge n_i^{\text{term}} \geq 1], \quad (8)$$

Points per 100 possessions were then estimated as a function of COVERED, the count of each role on the floor, controlled for talent, and fixed for team-season, opponent, game period, and score margin:

$$\text{PPP}_i = \beta \text{COVERED}_i + \sum_c a_c n_i^c + g \text{Talent}_i + \phi_{\text{ts}(i)} + \psi_{\text{opp}(i)} + \chi_{\text{per}(i)} + \omega_{\text{sb}(i)} + \varepsilon_i. \quad (9)$$

Including the category counts alongside COVERED matters, because COVERED is a nonlinear threshold built from those same counts: this lets β isolate the effect of completeness itself, beyond just having more of each role individually. The pre-registered hypothesis was $\beta > 0$, with a strict significance bar (team-season cluster-robust $p < 0.01$) and a bootstrap confidence interval excluding zero.

Hypothesis 1 (Coverage > accumulation, T_{cov}). $\beta > 0$ with team-season cluster-robust $p < 0.01$ and a cluster-bootstrap 95% CI excluding 0.

Result. Coverage does not predict scoring efficiency. $\hat{\beta} = +0.42$ points per 100 possessions, small and not statistically different from zero ($t = 0.53$, $p = 0.60$), with a confidence interval of $[-1.15, +1.90]$ across 0. This held up across several versions of the test (Table 5): removing the talent control, adding play-type controls, requiring stricter role classification, the estimate barely moved. The one term that reliably did predict more points was simply having more organizers on the floor, $+2.35$ points per 100 ($t = 4.62$), a real effect, but a separate and unsurprising one: more high-usage playmakers means more scoring. It has nothing to do with completeness.

Table 5. Efficiency boundary. Possession-level PPP, $N=4.2 \times 10^5$, 90 team-seasons. Coefficients in points per 100; CIs team-season cluster-bootstrap. COVERED is never distinguishable from zero; organizer accumulation is the robust driver.

Specification / term	pts/100	t	95% CI
COVERED: primary (M1)	+0.42	0.53	$[-1.15, +1.90]$
COVERED: with play-type FE	+0.44	0.53	$[-1.29, +2.04]$
COVERED: talent control removed	+0.43	0.54	$[-1.15, +1.94]$
COVERED: $\geq 4/5$ classified	+0.26	0.37	$[-1.23, +1.67]$
$n_{\text{organizer}}$ (accumulation)	+2.35	4.62	—

The sample gives enough precision to rule out a large scoring bonus from completeness, but not a small one, the confidence interval’s upper bound (+1.90) is not quite zero. Because coverage was designed to be separate from talent, and the estimate stayed small and stable across every version of the test, this is a limit, not a failed test.

Isolating shaper presence alone. A narrower version of this question was also tested: does simply having a shaper on the floor, regardless of whether the offense is otherwise “complete,” change scoring at all? COVERED was replaced with a simple indicator, $\text{HasShaper}_i = \mathbb{I}[n_i^{\text{shap}} \geq 1]$, keeping the linear role counts so the coefficient reflects shaper presence net of raw accumulation. The answer is the same: no. $\widehat{\text{HasShaper}} = +0.00$ points per 100 possessions ($t = 0.00$, $p = 1.00$) once everything else is controlled for. A related test, whether having more shapers on the floor at once helps, comes back marginal (+1.17 per 100, $p \approx 0.02$) but does not clear the pre-registered significance bar, so it is not treated as confirmed. Shapers change what an offense runs (Sec. 6.3), but not how many points it scores doing it.

What this leaves. Structural organization is measurable, and it is a real property of how an offense is wired, but it is not another way of predicting who scores efficiently. That is not a failure of the measure, it is what the measure actually is: a description of organization, not a hidden efficiency stat.

6.4.1 Exploratory shaper-tier heterogeneity

As a post-hoc check motivated by an out-of-sample 2025–26 review (Sec. 3), this paper splits SHAPERS at within-class median residual influence into hub vs. marginal, then splits marginal by position (big vs. wing). Replacing COVERED with three presence indicators on the same pre-registered three-season sample yields no tier that clears the paper’s 0.01 confirmatory bar (Table 9; detail in the appendix). The check does not revise Table 5 or the HasShaper null above.

6.4.2 Does the offense’s identity still change, even when scoring doesn’t?

Structural influence, $L(p)$, is a hypothetical measure. It answers a question that never actually happens on a basketball court: if a player’s initiations were erased from the entire season, how different would the offense’s fingerprint be? That is a clean mathematical experiment, but it leaves open a real question: does that hypothetical number correspond to anything that actually happens in real basketball, or is it a product of deleting rows from a dataset? If Gobert actually sits out real possessions, does Minnesota’s real, observed offense actually shift the way the hypothetical

calculation predicted it would? This section answers that directly, using real on/off data instead of a simulated removal. This provides an important check on the leave-one-out measure. If players with high hypothetical structural influence did not also show larger real on/off changes, it would raise a serious question about whether the leave-one-out calculation corresponds to anything observable on the court.

How this differs from the earlier calculation. Structural influence, $L(p)$, was calculated earlier as a hypothetical: delete a player’s stats entirely from the season and rebuild the fingerprint from scratch (Eq. 4). This test uses real data instead. For each player-season with adequate sample (at least 400 possessions both on and off the floor), the team’s actual possessions are split by whether that player was on the court, two separate standardized fingerprints are built, $\mathbf{A}_{\text{on}}$ and $\mathbf{A}_{\text{off}}$, from those real subsets, and the difference between them is measured:

$$D_A(p) = 1 - \cos(\text{vec}(\mathbf{A}_{\text{on}}), \text{vec}(\mathbf{A}_{\text{off}})). \quad (10)$$

Unlike $L(p)$, which is a hypothetical full-season removal, D_A is built entirely from real minutes the player actually sat out, real benchings and rest patterns, not a counterfactual.

Result. Shapers show meaningfully more identity change between on-floor and off-floor minutes than similar low-usage players who are not shapers. 96 shaper/non-shaper pairs were matched on nearly identical usage rates (average usage gap of just 0.008), and shapers showed a larger shift in 81% of those pairs, with a mean paired difference of $\Delta D_A = +0.115$ ($t = 6.35$, $p \approx 7 \times 10^{-9}$). The play-type mix itself also shifted more when a shaper sat, not just the fingerprint ($\Delta D_\pi = +0.011$, $p \approx 10^{-4}$).

More directly answering the question this section opened with: the hypothetical leave-one-out $L(p)$ correlates strongly with real D_A ($r = +0.60$ overall, $r = +0.79$ within shapers specifically). That’s important because the hypothetical measure and the real on/off measure are calculated in completely different ways. One asks what happens to the team’s fingerprint if you remove the player from the full-season data. The other looks at what actually happens when he is on and off the court. They line up.

Table 6. On/off identity displacement D_A . Eq. 10 by structural class. Not pre-registered. $N = 908$ rotation player-seasons with ≥ 400 possessions on and off. Organizers are a reference class; the licensed claim is the usage-matched shaper vs. role-occupant contrast in the text ($\Delta D_A = +0.115$, $n = 96$ pairs).

Class	n	mean D_A	median D_A	mean usage
Primary Organizer	96	0.214	0.170	0.244
SHAPER	102	0.202	0.151	0.144
Terminal Scorer	361	0.112	0.094	0.233
Role Occupant / Specialist	349	0.102	0.085	0.148

What this means. The leave-one-out framework from Section 4 is validated against real basketball, not just internally consistent math. Separately, even though a shaper’s presence does not move the scoreboard (Sec. 6.4), it does change what the offense actually looks like while he is on the floor, confirmed here with real, observed data rather than a hypothetical. The value of a shaper, based on this evidence, is not in points, it is in helping hold together the team’s actual identity and

structure. Because this test was not pre-registered, it is reported as a disclosed follow-up finding, not a confirmed result, and a clean replication on a new season would be the right next step to confirm it.

One open question this paper cannot answer: whether any opposing defenses actually detected and exploited this reorganization in real time. The defensive-attention hypothesis raised earlier (Sec. 4.3), that a defense can start treating a cut as a one-off rotation rather than a real threat once fewer players are running it, assumes defenses adjust to the new shape of the offense. That assumption is untested here. It is possible that opposing defenses, across these three seasons, did not specifically scout or exploit the exact reorganization a shaper’s absence produces, in which case the offense’s structure changed without any defense actually adjusting to it, which would be consistent with the null PPP result in Section 6.4. Whether NBA defenses identify and adjust enough to adapt to this kind of structural reorganization is an open question this paper’s data cannot answer, since it would require studying defensive coverage (which requires other data), not just offensive initiation patterns. This gap is a natural direction for the defense-side question raised in Section 7.

6.5 Is organizational function partly player-specific?

The portability analysis is a supporting check on whether the measure is real, not the paper’s primary claim. If a player’s high structural influence reflects something about the player himself, not just the old team around him, then his measured footprint should reappear after he changes teams. This section tests that directly across all movers, then examines the shaper category as a stress test, since it is the category where low usage most clearly hides structural dependence.

Design. Each mover in this test actually played for two teams. His origin signature captures what his removal would disrupt on his old team, calculated from real data before the trade. His destination signature captures what his removal would disrupt on his new team, calculated from real data after the trade, once he had actually joined it (Eq. 7). The question is whether those two disruption patterns look alike: does the same player create a similar structural gap on two completely different rosters? These two signatures are compared using cosine similarity, called self-similarity ($\text{self_sim}(p) = \cos(\Delta \mathbf{A}_p^{\text{orig}}, \Delta \mathbf{A}_p^{\text{dest}})$).

The obvious worry is that any apparent match could just be coincidence, maybe both teams happen to look similar for reasons that have nothing to do with him specifically. That is what the decoy comparison rules out. For each mover, the test also checks whether his origin signature predicts the destination team’s fingerprint better than a decoy would, a different player already on the new team whose own play-type profile is closest to his ($K = 5$ profile-matched decoys). If a random stylistic double would have predicted the new team’s fingerprint just as well, there is nothing player-specific going on, it is just “anyone who plays like this would create this pattern here.” Only when the real mover’s own signature beats his decoy’s does that count as evidence that something about him individually, not just his general style, is carrying over:

$$\Delta(p) = \text{self_sim}(p) - \text{decoy_sim}(p). \quad (11)$$

A positive Δ means the player’s own structural footprint predicts his destination team’s footprint better than a teammate who simply runs similar plays would. The pre-registered hypothesis was that the median Δ across all movers is greater than zero, tested with a strict permutation test and bootstrap confidence interval.

Hypothesis 2 (Portability, T1). $\text{median}_p \Delta(p) > 0$, with a matched-set (within-pool) exchangeability permutation $p < 0.01$ and a destination-franchise cluster-bootstrap 95% CI excluding 0.

Table 7. Portability by class. $n = 294$ movers; pre-registered three-season window, 2022–23 through 2024–25 (2025–26 excluded from confirmatory portability). Classes use the mover-sample usage median $\times r > 0$ cut matching Eq. 6. Δ is the profile-matched signature excess (Eq. 11); “portable” is $\Delta > 0$. Kruskal–Wallis across all four classes: $H = 3.85$, $p = 0.28$ – medians and %-portable are reported as descriptive statistics only; the omnibus test does not support treating this as a confirmed class ranking.

Class	n	Median Δ	% portable
Primary Organizer	30	+0.134	70.0%
SHAPER	24	+0.124	58.3%
Terminal Scorer	117	+0.083	60.7%
Role Occupant / Special.	123	+0.018	54.5%

Feasibility check. 294 movers were confirmed across 30 destination franchises. Critically, a player’s displacement signature and his play-type profile are essentially uncorrelated with each other (Spearman $\rho = 0.007$), meaning a positive Δ cannot be waved away as “well, he just runs the same plays everywhere.” Movers were also carefully ordered by actual game date rather than season label, since a mid-season trade can leave two different stints carrying the same season label, which would otherwise risk mismatching stints across a season boundary.

Result: a player-specific component holds up, on average. Across all 294 movers, median self-similarity was +0.129 versus median decoy-similarity of +0.049, an excess of +0.056. All three pre-registered criteria passed. This held up under permutation testing ($p = 0.005$) and bootstrap confidence intervals that exclude zero (destination-franchise cluster-bootstrap 95% CI [+0.019, +0.086]). In plain terms: 58.8% of movers showed positive portability, versus the roughly 50% one would expect if nothing player-specific carried over at all. A real, player-associated piece of organizational function does persist across team changes, but imperfectly, the effect is real but modest, part player, part context. Because players are not randomly assigned to teams, this cannot be treated as proof of causation, just a real, robust association.

Differences by category exist, but aren’t reliable at this sample size. By category (Table 7), median Δ was +0.134 for organizers, +0.124 for shapers, +0.083 for terminal scorers, and +0.018 for role occupants. But the overall test comparing all four groups does not clear significance (Kruskal–Wallis $H = 3.85$, $p = 0.28$), so with only 24 shaper movers, this paper cannot confidently say organizers travel better than shapers, or shapers travel better than terminal scorers. These four numbers are reported as descriptive, not as a confirmed ranking. The rest of this section and Section 6.6 stay focused on shapers specifically, examining that category on its own terms rather than in comparison to the others.

Shaper portability is a distribution, not one number. Among the 24 shaper movers, three cases anchor the range: Daniel Gafford near the top, Herbert Jones Jr. near the median, and Delon Wright near the bottom (Fig. 5; Table 8). These are not career grades, they are illustrations of what a strong, typical, and weak case of structural portability actually looks like.

The portability test asks whether a player’s removal disrupts the same relationships across two different teams, not whether he literally reproduces the same raw stat line on a new roster. A concordance map like Figure 6 shows, cell by cell, whether the origin and destination teams agree

on the direction of change for each play-type pair when the player is removed: consistent weakening colored one way, consistent strengthening the other, and no agreement left uncolored.

Gafford’s map is sparse but deep: two strong, consistent cells on cut-linked pairs dominate his footprint (self-similarity $+0.86$, $\Delta = +0.60$). In basketball terms, wherever Gafford goes, he creates the same specific structural gap when he sits, both in Washington and in Dallas, removing him weakens the same cut-linked pairs, in the same direction, on two rosters with entirely different personnel. That is a role he genuinely brings with him, not something his old team happened to be using him for.

Jones Jr.’s map is more scattered, some agreement, some disagreement, a weaker overall signal (self-similarity $+0.21$, $\Delta = +0.17$). His removal shifts some of the same pairs in the same direction on both teams, real signal, but it is mixed in with pairs moving in opposite directions and others with no consistent pattern at all. His role partially travels, but a meaningful part of what he did structurally looks specific to what his original team needed from him.

Wright’s map shows almost no agreement at all between his two teams (self-similarity -0.66 , $\Delta = -0.71$). Whatever structural gap his absence created in Washington does not match the gap his absence created in Miami. His structural role was not really his own, it was mostly shaped by what his original team specifically had him doing, and it did not reappear in a new context.

The lesson: a small number of rock-solid, repeatable structural connections, like Gafford’s cut-linked pairs, matter more than a larger number of loose, inconsistent ones. A few things a player reliably does everywhere predict his real structural value better than a broad footprint that only partially holds up once the roster around him changes. (Gafford’s origin window here is his partial pre-trade WAS 2023–24 stint, not the full WAS 2022–23 season in Table 3.)

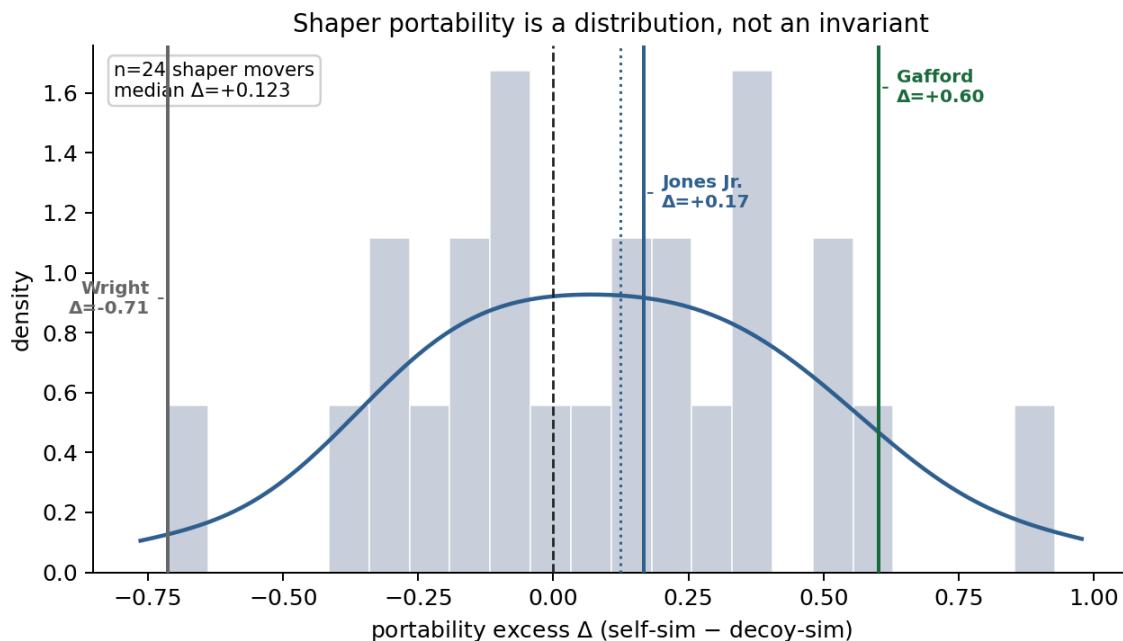


Figure 5. Portability is a distribution, not an invariant. Realized portability Δ among SHAPER movers ($n = 24$) in the confirmatory three-season sample (2022–23 through 2024–25; 2025–26 excluded). Positive Δ means the player’s origin structural footprint predicts his destination footprint better than a profile-matched decoy’s. Endpoints illustrate the distribution; they are not the story.

Table 8. Shaper portability endpoints. Confirmatory sample (2022–23–2024–25 only; 2025–26 excluded). Illustrations, not evidence. u = initiation share; L = origin→destination influence; Δ = Eq. 11.

	Strong: Gafford	Median: Jones Jr.	Weak: Wright
Init. share u	0.036	0.030	0.016
L (origin)	0.335	0.025	0.030
L (dest.)	0.007	0.014	0.002
self-sim	+0.86	+0.21	−0.66
Δ	+0.60	+0.17	−0.71
Portable	Yes	Yes	No

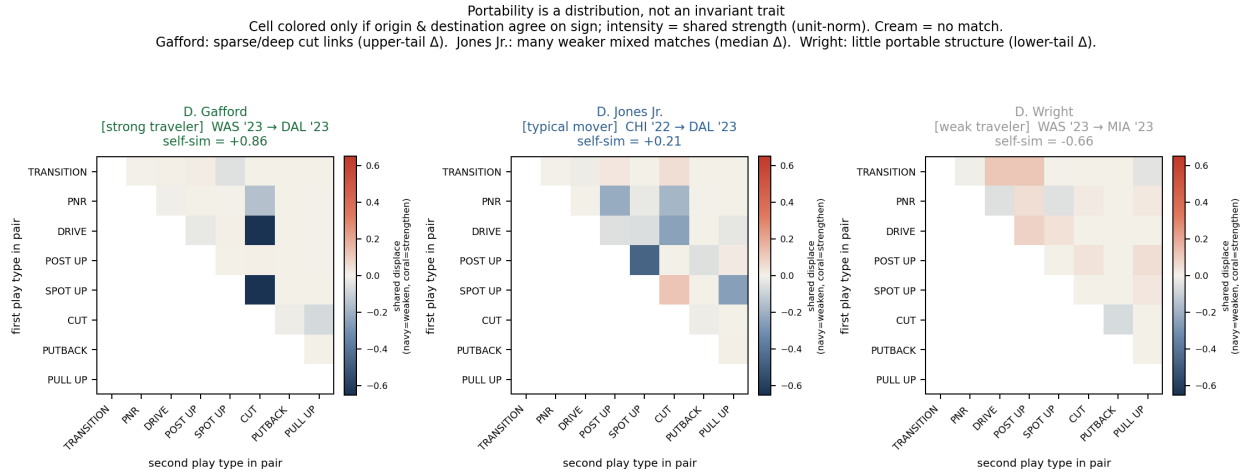


Figure 6. Depth of concordance, not a success/failure binary. Confirmatory three-season movers only (2022–23 through 2024–25). Each panel is a concordance of origin and destination displacement signatures (Eq. 7), not two raw heatmaps side by side. A cell is colored only when both teams agree on the sign of that play-type-pair shift when the player is removed: navy = shared weakening of the link, coral = shared strengthening, cream = no same-sign match. Intensity is shared strength after unit-normalizing each signature. Self-sim (panel title) is the cosine of the full signatures; it (not cell count) ranks the panels. Sparse/deep concordance (Gafford) outranks many weaker mixed-sign matches (Jones Jr.); Wright shows little portable structure. The figure shows whether the same player’s removal disrupts similar relationships across contexts, not career outcomes.

Does landing on the right team explain the variation? The obvious front-office explanation for why portability varies is fit, maybe a player’s role reappears only if his new team has room for it, and fails if someone else already fills that role. This is tested using a fit score, essentially the

inverse of how closely his new teammates already resemble him structurally:

$$\text{fit_score}(p) = -\text{decoy_sim}(p). \quad (12)$$

Result: fit doesn’t explain it. Among shaper movers, fit score barely correlates with portability at all (roughly -0.09 to -0.39 depending on the measure). Landing on a team that structurally has a hole for a shaper’s role does not reliably predict whether that role actually travels. This is a precise null on this specific measure of fit, not a claim that scheme fit never matters at all in any sense.

6.6 Can organizational function transfer be predicted?

Section 6.5 established that portability is real but variable, and that destination fit does not explain the variation. This section asks a narrower question: is there anything about a shaper’s own game, before the trade, that predicts whether his structural role will travel with him?

The short answer is no, not reliably. A composite score built from 63 different features of the origin and destination context was tested, and none of it survives correction for testing that many things at once (Appendix C). The single most promising lead is a player’s own cut share before the trade: shapers who ran a lot of cuts on their original team were more likely to be portable (75% portable among high-cut shapers, versus 37.5% among low-cut shapers), but with only 24 shaper movers, this does not reach statistical significance ($p = 0.13$). It is reported as a directional and unconfirmed finding, not a prediction tool.

7 Discussion

The framework is the main contribution, not any one result. This paper starts with a fairly simple idea: NBA teams clearly organize offensive responsibility in ways that don’t show up in ordinary player statistics, but there has not been a straightforward way to measure that organization. The graph framework provides one way to do that. The results in Section 6 are the first test of what the measure picks up, where it seems to correspond to real basketball, and where it does not.

The first set of results gives the measure some support, but also puts clear limits around it. Structural influence is not just another value metric: once usage is removed, it has essentially no relationship with net rating. SHAPERS also correspond to something observable on the court, shifting the team’s play mix even though they themselves are low-usage players. And when players change teams, part of their structural footprint follows them, although imperfectly. At the same time, the framework does not predict an efficiency advantage. That distinction matters: this appears to be measuring how an offense is organized, not how good it is.

The efficiency null (Sec. 6.4) is a boundary, not a failed test. A measurement of organizational structure does not need to reproduce the outcomes used to evaluate players: structural influence is measurable, but it is not the same as offensive efficiency. The disclosed on/off identity test (Sec. 6.4.2) then supplies the follow-up: shaper absence displaces the observed association fingerprint more than usage-matched role-occupant absence. Scoring efficiency does not move, but identity does. That is the scoped “so what” for the shaper class in this paper: organizational continuity under substitution, not a PPP effect.

The null efficiency result is still useful because it tells us what structural influence is not. Losing a shaper changes the observed organization of the offense, but this paper does not find evidence that the change itself produces a measurable scoring penalty. Organizers are different in one important

respect: having more of them on the floor is associated with higher scoring efficiency. But that is a separate result and is not evidence that organizational completeness itself improves efficiency. The simplest interpretation is that organizational dependence and offensive value are not the same thing.

What becomes measurable now: from players to teams. Structural influence identifies who an offense’s current wiring depends on. It does not identify who the offense cannot afford to lose. The framework is not only about individuals. Structural influence is a team-level distribution, not just a player-level score, and its concentration across a roster remains a natural candidate for organizational resilience at the team level. Two offenses can score identically and still differ in how much they depend on one or two load-bearing players. The player-level on/off identity result (Sec. 6.4.2) is evidence that this dependence is real under substitution. Elevating roster influence concentration to a confirmatory team-brittleness claim is still future work.

With a reproducible measure of offensive wiring, researchers and practitioners can ask questions that previously had no quantitative handle, at both the player and team level:

- Which teams are structurally brittle, and to whose absence specifically?
- Which injury would force the largest offensive reorganization?
- Which player replacement requires the least rewiring, holding scoring fixed?
- Does a given trade preserve or fundamentally rewire offensive identity?
- How do coaching systems differ in structural dependence, not only in outcomes?
- How do rookies assume organizational responsibility over time?
- How does organizational structure evolve across seasons?
- Which defenders most disrupt an opponent’s wiring (the inverse perturbation question flagged below)?

These questions can still matter even if structural influence is not a value metric. A team does not need to know that a player is worth a fixed number of extra points to ask whether losing him will force a major change in how the offense operates. Identifying where that dependence exists could be useful for injuries, roster construction, trades, and player replacement. The results in this paper are a first test of the framework, not a complete list of what it can eventually be used for.

What remains outside the possession graph. Portability is unpredictable given what a possession-initiation graph can see. Other channels include coaching deployment, spatial fit, defensive scheme, specific on-court partnerships, and signature noise from low initiation volume. An exploratory panel on “when to acquire” (brittleness / multiplier triggers) does not find a timing rule and reinforces the efficiency boundary (Sec. 6.4).

Future work: the inverse question on defense. Everything above is an offensive wiring diagram. The graph also carries defensive on-court lineups, which opens a question this paper does not answer: which defenders reshape the opponent’s offensive wiring the most. That requires its own construct-validity work before it can be evaluated with the same discipline applied to portability. Separately, a defense can already adjust coverage, help, and matchups in anticipation of the known post-removal play-mix pivot documented in Sec. 6.3 (more cuts and pick-and-rolls, fewer spot-ups and drives). That use of the framework does not require a new defender-attribution model.

8 Limitations

- **Initiation only.** Edges record who initiates each play type, not the full pass/screen sequence. Intra-possession chains are outside the data.
- **No player tracking.** No spatial data. Organizational function is inferred from initiation alone, which helps reproducibility but limits what can be said about mechanism. As an illustration: Draymond Green, commonly cited as an archetypal offensive shaper, categorizes as `ROLE OCCUPANT / SPECIALIST` under Eq. (6) in all three sample seasons, since his organizational contribution is understood to operate primarily through passing and communication, both outside what this graph represents.
- **Association only.** Rosters are chosen, not randomized. Portability reflects that wiring ability tends to reappear, and origin-side selection is not fully excluded.
- **Small-influence estimation noise.** Near $r = 0$, taxonomy boundaries are approximate for very low-initiation players.
- **Efficiency as a boundary.** The coverage null (Sec. 6.4) marks a boundary of the instrument, not a failure of it.
- **On/off identity test is post-hoc.** Sec. 6.4.2 was not pre-registered. It is a follow-up to the efficiency boundary using a different outcome (D_A , observed on/off association displacement) from leave-one-out $L(p)$. A prospective replication remains the gold-standard check.
- **$L(p)$ and D_A use slightly different possession universes.** Primary $L(p)$ is computed on the full initiation-edge universe. Primary D_A uses the on-court five-player filtered pool (Sec. 6.4.2). A sensitivity check aligning both on the initiation-edge universe yields $r = +0.51$ ($n = 908$; `dA_L_common_universe_check.py`, `output/dA_L_common_universe_check.csv`).
- **Backward-looking by construction.** Origin-side screens extrapolate from what a player has been, not developmental upside.
- **Cut/drive screen is exploratory.** Directional at $n = 24$ shaper movers, not part of the pre-registered portability protocol (Sec. 6.6). A prospective blind battery on a new season remains the gold-standard confirmation. 2025–26 was deliberately excluded from confirmatory tests.
- **Large residual outside the initiation graph.** Most within-shaper Δ variation is unexplained after cut/drive. Destination context is FDR-null.
- **Fit operationalization.** Destination fit is `-decoy_sim` (Eq. 12), which captures structural vacancy rather than scheme fit or coaching trust.
- **Sample and era.** Three seasons, on-court analyses restricted to the tracking subset. Median-split thresholds may shift across eras (appendix).

9 Conclusion

Most basketball analytics measure what players produce. This paper introduces a reproducible graph framework for measuring how they organize production. Offenses are wiring diagrams, team-level maps of who initiates which play types and whether the same players jointly hold multiple jobs. Structural influence quantifies each player’s contribution to that organization via network perturbation.

The main contribution is not a new value metric for identifying hidden stars. It is a way to quantify one part of offensive organization that has mostly been discussed qualitatively. The first application here suggests that this structure is distinct from conventional value measures, visible in real changes to offensive organization, and partly portable across teams. The framework also leaves a number of questions open, particularly around team brittleness, injuries, roster replacement, and defensive disruption.

Code Availability

Analysis code and paper materials are available at https://github.com/ramizheman/Basketball_Structural_Influence.

AI Disclosure

Large language models (Claude, Anthropic; Grok, xAI) were used to help develop the mathematical formulation, write the analysis code, and draft this paper’s text. The author directed the research question, supplied and validated the data, and reviewed and edited all results and text.

A Appendix

A. Threshold sensitivity. Role assignments (Eq. 6) under 25th/75th-percentile splits, with class-membership stability against the median split.

B. Structural-influence robustness. Frobenius-distance influence as an alternative to cosine (Eq. 4); within-player influence coefficient of variation across seasons as a stability check.

C. Portability robustness. Origin- vs. destination-clustered bootstraps, decoy count K at its pre-registered value, and the permutation null construction (Sec. 6.5).

Cut/drive travel-screen battery (detail for Sec. 6.6). Beyond the main-text headline, the origin-side screen was checked for robustness as follows. Within-class $\text{corr}(\text{cut}, \Delta)$ is +0.32 for shapers vs. +0.12 (organizers), +0.06 (terminals), +0.10 (role occupants), though the shaper estimate itself is not individually significant. Partial correlations controlling for signature concentration/norm, initiation share, and origin influence/usage jointly are consistent in sign and similar in magnitude to the raw estimate ($r = +0.34$ to $+0.39$, $p = 0.06$ – 0.11), and a one-move-per-player bootstrap (removing duplicate-mover clustering) keeps $\rho > 0$ in 100% of 2,000 resamples (median $\rho = +0.39$). Dropping the top-three Δ movers weakens the estimate further ($\rho = +0.22$, $p = 0.33$, $\text{AUC} = 0.70$), and only one of two temporal halves clears a minimum sample size. Across 63 destination-side features, no signal survives Benjamini–Hochberg FDR correction at $q < 0.05$ (a small number of raw $p < 0.05$ hits are consistent with chance). Descriptive (not a second classifier): high-cut portable cells include Hayes (37% cut), Eubanks (37%), Metu (32%), and Mamukelashvili (22%); drive-heavier non-portable contrasts include Delon Wright (Sec. 6.5 lower-tail endpoint) and Evan Fournier (16% drive, $\Delta = -0.16$). Cut diet is a screen, not a sufficient condition. Relative to the pre-registered portability test, the cut/drive pattern is exploratory, modest ($\sim 10\%$ of within-shaper rank variation in Δ), and not statistically confirmed at $n = 24$.

D. Behavioral analyses. Full within-team play-mix deltas for all four classes and the route-entropy construction and null (Sec. 6.3).

E. Coverage pre-registration. The complete pre-registered protocol for the efficiency boundary (Sec. 6.4): feasibility gate, frozen talent control, fixed-effects specification, and binding no-fallback stop conditions.

F. Structure-only negative analyses. Additional structure→efficiency attempts (absence-impact, structural replacement, concentration-vs-defense), each reported as a registered negative and each reinforcing the efficiency boundary of Section 6.4.

Exploratory shaper-tier heterogeneity (detail for Sec. 6.4.1). As a post-hoc check (motivated by an out-of-sample 2025–26 review; Sec. 3), this paper splits SHAPER at within-class median residual influence into hub vs. marginal, then splits marginal by position (big vs. wing). Replacing COVERED with three presence indicators (HASHUB, HASMARGBIG, HASMARGWING) on the same pre-registered three-season sample yields Table 9: *none* of the three tiers clears the paper’s 0.01 confirmatory bar. The check is exploratory relative to the pre-registered efficiency tests and does not revise Table 5 or the HASSHAPER null in Sec. 6.4.

Table 9. Exploratory shaper-tier PPP effects. $N=420,993$, 90 team-seasons. Coefficients in points per 100; confirmatory bar $p < 0.01$ as reporting benchmark only.

Tier	pts/100	p	Confirmatory bar ($p < 0.01$)
HasHub	+1.10	0.077	not met
HasMargBig	+1.70	0.025	not met
HasMargWing	+1.29	0.122	not met

G. Shaper roster (2025–26). Table 10 lists every SHAPER player-season in the 2025–26 rotation pool (Sec. 6.1), with the same usage, influence, MPG, and PPG fields reported in aggregate in Table 2. This is a roster, not a ranking: it is provided for inspection and replication, not as an endorsement of any individual player’s value.

Table 10. 2025–26 Shaper roster ($n = 41$). Exploratory addendum only; not part of the confirmatory three-season sample. Usage is box-score usage rate; influence is $L(p)$ (Eq. 4); MPG and PPG are box-score minutes and points per game. Sorted by team, then influence (descending).

Player	Team	Usage	Influence	MPG	PPG
<i>BKN</i>					
N. Claxton	BKN	0.175	0.061	27.8	11.7
J. Wilson	BKN	0.177	0.017	15.9	6.4
<i>BOS</i>					
N. Queta	BOS	0.145	0.170	25.3	10.2
S. Hauser	BOS	0.141	0.022	24.8	9.2
<i>CHA</i>					

Continued on next page

Table 10 , continued

Player	Team	Usage	Influence	MPG	PPG
R. Kalkbrenner	CHA	0.113	0.020	21.4	7.6
M. Diabate	CHA	0.112	0.014	26.0	7.9
<i>DAL</i>					
M. Bagley III	DAL	0.180	0.022	20.0	10.5
D. Gafford	DAL	0.151	0.016	21.7	9.5
<i>DEN</i>					
T. Hardaway Jr.	DEN	0.177	0.036	26.6	13.5
C. Braun	DEN	0.142	0.022	31.8	12.0
C. Johnson	DEN	0.144	0.020	30.5	12.2
S. Jones	DEN	0.097	0.016	22.1	5.5
J. Pickett	DEN	0.148	0.014	16.1	5.2
<i>DET</i>					
D. Robinson	DET	0.152	0.044	27.4	12.2
I. Stewart	DET	0.164	0.017	22.7	10.0
J. Green	DET	0.143	0.015	17.6	6.9
<i>GSW</i>					
G. Payton II	GSW	0.177	0.033	15.6	7.5
W. Richard	GSW	0.123	0.019	20.0	6.4
A. Horford	GSW	0.163	0.016	21.5	8.3
<i>LAC</i>					
B. Lopez	LAC	0.171	0.020	21.8	8.5
<i>LAL</i>					
J. Hayes	LAL	0.124	0.025	18.3	7.5
<i>MEM</i>					
K. Caldwell-Pope	MEM	0.173	0.034	21.3	8.4
<i>MIA</i>					
K. Ware	MIA	0.174	0.352	22.1	11.1
<i>MIL</i>					
J. Sims	MIL	0.095	0.139	19.7	5.0
A. Green	MIL	0.135	0.025	29.1	10.4
<i>MIN</i>					
R. Gobert	MIN	0.127	0.059	31.3	10.9
<i>NOP</i>					
Y. Missi	NOP	0.122	0.019	19.7	5.7
<i>NYK</i>					
M. Robinson	NYK	0.102	0.233	19.6	5.7
<i>OKC</i>					
I. Hartenstein	OKC	0.153	0.125	24.2	9.2
I. Joe	OKC	0.178	0.022	21.2	11.1
A. Caruso	OKC	0.152	0.015	18.2	6.2
<i>ORL</i>					
G. Bitadze	ORL	0.128	0.269	15.2	5.9
J. Carter	ORL	0.166	0.012	16.4	6.4
<i>PHI</i>					

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Table 10 , continued

Player	Team	Usage	Influence	MPG	PPG
A. Bona	PHI	0.108	0.061	17.4	4.8
A. Drummond <i>PHX</i>	PHI	0.143	0.013	19.5	6.4
M. Williams <i>POR</i>	PHX	0.167	0.015	23.6	11.7
R. Williams III	POR	0.129	0.067	17.1	6.7
D. Clingan <i>SAC</i>	POR	0.163	0.038	27.2	12.1
D. Cardwell <i>SAS</i>	SAC	0.108	0.046	20.6	5.4
L. Kornet <i>TOR</i>	SAS	0.100	0.103	21.0	6.5
J. Poeltl	TOR	0.150	0.037	25.0	10.7

H. Play-type inferrer from public play-by-play. No Synergy, Second Spectrum, or tracking feed is used. Possession labels are inferred from NBA Stats play-by-play V3 fields available via the public API, principally `actionType/subType`, event description text, assist tokens on made shots, shot distance, and court coordinates, after events are normalized into a possession chain. A rule-based classifier maps those signals onto nine labels (`TRANSITION`, `PNR`, `DRIVE`, `POST_UP`, `SPOT_UP`, `CUT`, `PUTBACK`, `PULL_UP`, `OTHER`); association and structural-influence calculations use the eight initiating types listed in Sec. 3, with residual `OTHER` possessions excluded from the canonical set after labeling (and after the foul- and PNR-calibration passes below).

Rules fire in fixed priority (highest first). Table 11 summarizes the main PBP→label mappings; confidence scores accompany each rule so downstream gates can prefer high-signal labels when needed.

Table 11. Possession classifier signals. Public PBP signals to initiating play type. PNR rows use assist flags that appear only on made shots unless noted.

PBP signal (subType / text / assist)	Label
Putback / tip subtypes	PUTBACK
Cutting / alley-oop subtypes	CUT
Fadeaway, hook, turnaround (or post text)	POST_UP
Running / fast-break subtypes or text	TRANSITION
Assisted pull-up or floater (made)	PNR (ball handler)
Assisted rim finish ≤ 6 ft, not cut/running (made)	PNR (roll man)
Missed unassisted pull-up outside the paint	PNR (low confidence)
Driving subtype, no PNR cue	DRIVE
Unassisted pull-up / step-back	PULL_UP
Assisted 3PT with no stronger creation cue	SPOT_UP
Foul-ending possession left as <code>OTHER</code>	player-tendency relabel

The strings “pick” and “screen” never appear in V3 descriptions, so PNR is never read off a ball-screen token. It is inferred from assist-plus-subtype/distance rules on makes, a low-confidence missed-pull-up correction, and the player-level calibration pass summarized in Sec. 3. Without calibration, the description-based inferrer alone recovers PNR initiations at roughly a quarter of

their true rate (raw inferrer output $\approx 4\%$ of possessions vs. an expected $\approx 20\text{--}25\%$ from play-type tracking proxies). The shot-event proxy is used only to set season-level promotion *targets*; it does not enter the association matrix or structural influence directly. Those objects use the final calibrated possession labels (inferred-native PNR plus promoted `PULL_UP`/`OTHER` possessions). Promotion targets come from a season-level PNR-rate proxy built from NBA play-by-play shot events (shot subtype, distance, and assist flags on makes), scaled by a fixed factor of 2.5 from shot-level to possession-level rates and capped at 33% of a team’s half-court possessions. Each candidate’s promotion score is that player’s own historical inferrer-only PNR rate (team rate if under 10 tagged possessions) times a fixed prior by current label and confidence (`PULL_UP`: 0.55/0.35/0.15 for conf $< 0.45/0.45\text{--}0.65/> 0.65$; `OTHER` at conf 0.30: 0.25), so promotion is player-level rather than a flat team-wide adjustment. A further per-game cap limits drain of any one team’s `PULL_UP` bucket to at most half of that bucket; remaining promotions come from `OTHER`, and the pass stops rather than overshoot if both are exhausted. Every promoted possession retains its pre-calibration label and confidence (enabling exact rollback) and is stamped at confidence 0.22, below any inferrer-native PNR label. Applied this way across seasons 2022–23 through 2025–26 (the fourth season only for exploratory addenda), seasons land within a point of each other on PNR share (18.3–20.0%).

Foul treatment. Possessions that end in free throws without a classified shot action often fall through to `OTHER` at low confidence. A foul-tendency pass then relabels those `OTHER`+foul-ending possessions using the fouled initiator’s season primary action type (drive, post-up, PNR, cut, spot-up, pull-up, or transition), stamped at mid confidence and marked `foul_tendency_relabeled`. Free-throw-only trips that are not attributable to a half-court initiation are excluded from half-court efficiency tallies where PPP is computed. A backward attribution audit recovered 98.6% of remaining foul-ending `OTHER` possessions by linking them to the initiating offensive action; correcting misclassified foul endings shifted play-type PPP by at most 0.007 points and produced no play-type ranking changes, so the association-matrix corpus inherits the same labels without a material sensitivity to that residual.

I. Play-type pair glossary. Grouped by family, each with the on-court meaning and what a defense loses if that pairing disappears. Pairs with weak or no natural basketball reading are marked as such rather than forced into an explanation.

Ball-screen family.

- **PNR–DRIVE:** a ball handler who both runs pick-and-roll and drives without a screen. High overlap means he is live in ball-screen and isolation actions alike: a defender cannot relax off ball-screen coverage expecting he only operates off a pick.
- **PNR–POST_UP:** a big who both rolls/pops off ball screens and posts up. High overlap signals an interior hub who can be fed straight or used in an action, forcing bigger decisions from help defenders.
- **PNR–SPOT_UP:** a player who runs pick-and-roll and also spaces off it. Common for a secondary ball handler who initiates the screen action then relocates to shoot.
- **PNR–CUT:** covered in the main text (Gobert’s own dual threat: rim runs and roll starts).
- **PNR–PUTBACK:** weak natural reading. Likely reflects a big who is live in ball screens and also crashes the offensive glass, not a designed two-way action.

- **PNR–PULL_UP:** a ball handler who runs pick-and-roll and also creates his own pull-up jumper off the dribble without one. High overlap describes a primary shot creator, hard to funnel into one action.

Drive-adjacent.

- **DRIVE–POST_UP:** a player who both attacks off the dribble and posts up, usually a wing or forward who can operate from the perimeter or the block. Removes a defender’s ability to guard him with one strategy.
- **DRIVE–SPOT_UP:** driver who is also live standing still. Sag off the drive and he shoots; crowd the catch and he drives.
- **DRIVE–CUT:** covered in the main text: driver who is also live cutting off the ball when he is not initiating.
- **DRIVE–PUTBACK:** weak natural reading, mostly reflects a player active around the rim generally rather than one designed action.
- **DRIVE–PULL_UP:** a player live both attacking downhill and pulling up mid-drive. Standard shot-creator combination, hard to defend with a single coverage.

Post-adjacent.

- **POST_UP–SPOT_UP:** the inside-out combination described in the MIN/HOU comparison, a player who posts up and also spaces to shoot. Forces a defense to choose between fronting the post and closing out on the catch.
- **POST_UP–CUT:** a big who posts up and also cuts when not posting. Common combination for a traditional low-post player who also moves without the ball.
- **POST_UP–PUTBACK:** natural pairing for a true post player who also crashes the glass off his own or others’ misses.
- **POST_UP–PULL_UP:** weak natural reading. Rare for the same player to both post up and create a perimeter pull-up at real volume.

Cut-adjacent (beyond those above).

- **CUT–SPOT_UP:** covered in the main text: cut-and-replace action (cut to the rim, relocate if it is not there).
- **CUT–PUTBACK:** a player live around the basket generally, cutting and crashing for offensive boards. Common for a low-usage big.
- **CUT–PULL_UP:** covered in the main text: an unusual combination since cutters and pull-up shooters are typically different player types; when it appears, it signals a versatile wing.

Spot-up-adjacent (beyond those above).

- **SPOT_UP–PUTBACK:** weak natural reading, likely reflects a floor-spacing big who also crashes the glass, not a linked action.
- **SPOT_UP–PULL_UP:** a player live catching and shooting and also creating his own look off the dribble. Standard high-value shooter combination.

Transition-adjacent.

- **TRANSITION-PNR, TRANSITION-DRIVE, TRANSITION-POST_UP, TRANSITION-SPOT_UP, TRANSITION-CUT, TRANSITION-PUTBACK, TRANSITION-PULL_UP:**

transition largely reflects who is simply first up the floor or gets the ball in early offense, and tends to correlate with general offensive involvement rather than a specific tactical linkage. These pairs are included for completeness but generally carry weaker basketball meaning than half-court combinations; treat elevated values here cautiously and check whether they are being driven by a player's overall usage rather than a genuine two-way half-court threat.

Putback-adjacent (beyond those above).

- **PUTBACK-PULL_UP:** weak natural reading, an unusual combination; a high joint value here is more likely noise or a small-sample artifact than a real linked action.

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